

Article

Autonomous Drone-on-Drone Interception Using an Integrated LiDAR–Vision Detection System for High-Precision Capture

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Highlights

What are the main findings?

- The system successfully completed multiple autonomous real-world interception missions, achieving reliable target detection and following at ranges of up to 60 m, laying the foundation for successfully intercepting target UAVs.
- The dual-sensor approach with the integrated onboard LiDAR and camera system achieves a target detection accuracy of less than 0.4 m during dynamic interception scenarios, utilizing a custom-trained YOLO neural network for vision-based verification.

What are the implications of the main findings?

- The presented approach successfully addresses the critical precision limitations of ground-based sensors, therefore enabling autonomous net-based capture operations, reliably working in real-world scenarios.
- The developed system with a net-based UAV interception approach provides a safe counter-UAV method that preserves the target drone for forensic analysis while actively minimizing collateral damage risks.

Abstract

The rapidly increasing availability of low-cost commercial UAVs poses significant security challenges for critical infrastructure and law enforcement agencies. This paper presents an integrated LiDAR-based detection and vision-based verification system for an autonomous drone-on-drone aerial interception system. To eliminate the threat of possible dangerous target drones, the interception UAVs presented in this paper use a net to capture them safely in the air. The system addresses the critical limitation of ground-based sensors, which provide insufficient precision for reliable net-based capture operations. Moving beyond simulation-only approaches, the core novelty of this work lies in the successful real-world integration of these sensors on a strictly constrained aerial platform in size, weight and power to achieve sub-meter terminal guidance precision. The developed system uses real-time point cloud processing, DBSCAN clustering, and Moving Horizon Estimation tracking for the detection and tracking of the target. Vision-based verification uses a custom-trained YOLO neural network and achieves over 90% detection rates. The evaluation demonstrates a detection accuracy of less than 0.4 m at ranges exceeding 40 m during dynamic interception scenarios using RTK-GNSS ground truth. The dual-sensor



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approach successfully completed multiple autonomous interception missions with target detection ranges of up to 60 m, validating the capability of the system for safe, autonomous civilian UAV interception.

Keywords: UAV; drone interception; drone defense; LiDAR; computer vision; multi-sensor fusion

1. Introduction

In recent years, Unmanned Aerial Vehicles (UAVs) have transitioned from specialized research applications to widespread commercial and civilian use. This increase in easily accessible, low-cost commercial drones presents unprecedented security challenges for law enforcement agencies and operators of critical infrastructure. For example, recent security reports show the threat posed by UAVs, with the U.S. Department of Justice Office of the Inspector General reporting that drone incidents at federal prisons have increased strongly [1].

The security implications of widespread drone use have prompted significant research into counter-UAV technologies. While various approaches exist, including electronic jamming and kinetic systems, net-based capture methods offer the advantage of preserving the target drone for forensic analysis while minimizing collateral damage risks [2,3].

The Micro Drone Protection System (MIDRAS) project (2017–2020) established the foundational concept of using cooperative UAVs to capture intruding drones through net-based interception [4]. The MIDRAS project successfully demonstrated autonomous aerial interception using two UAVs in formation flight [5]. However, a critical limitation was the absence of onboard target detection capabilities, requiring the interceptor drones to rely entirely on external sensor data for target localization, which proved insufficient for precise terminal guidance and successful capture.

The follow-up project Innovatives Drohnenabwehrsystem (Engl. Innovative Drone Defense System) (IDAS) [6] was initiated to address these limitations and advance the technology toward operational deployment. Central to this advancement was the development of autonomous target detection and verification systems integrated directly onto the interceptor platform.

This paper presents the design, implementation, and evaluation of a LiDAR-based detection and vision-based verification system that enables autonomous guidance for the safe interception of civilian UAVs.

The enhanced system integrates a LIVOX mid-70 LiDAR sensor with a global shutter camera and neural network-based target verification, mounted on a redesigned octocopter platform. Rather than focusing solely on theoretical algorithmic development, the primary novelty of this work lies in applied systems engineering: successfully achieving sub-meter terminal guidance precision required for physical drone interception under strict constraints in real-world scenarios. Moving beyond simulation-based models, the key technical contributions include: (1) the integration of point cloud processing algorithms for dynamic target detection in cluttered environments, (2) vision-based target verification using trained neural networks for false positive reduction, and (3) extensive empirical evaluation of the fully autonomous catching system in real-world flight scenarios.

Experimental validation demonstrates a target detection accuracy of less than 0.5 m at ranges exceeding 50 m, representing a significant improvement in the precision of the air-to-air detection compared to external sensor-only approaches. The system successfully completed multiple autonomous interception missions, safely catching targets at various

distances. Figure 1 shows the autonomous catch of a target UAV using the proposed detection system.



Figure 1. Real-world autonomous catch of a target UAV.

2. State of the Art

The security threats posed by unauthorized civilian UAVs, such as camera-equipped quadcopters, have significantly increased the research interest in counter-drone technologies. The methods for detecting and neutralizing can be broadly categorized into three approaches: *Take-down methods* use kinetic or electronic impact to take down the flying object. These include net throwers [7], signal jammers targeting communication or GNSS links [8], and directed energy weapons used primarily in military applications [9]. These methods are often effective but risk uncontrolled crashes and therefore potential collateral damage. *Complete takeover methods* attempt to gain control of the target by signal hacking or GNSS spoofing [10]. These methods still pose significant security challenges in reality and can only be used against known UAVs, i.e., DJI [11]. *UAV catching methods* focus on controlled capture without causing uncontrolled crashes. These systems employ UAV-based net deployment, either through net launchers mounted on interceptor drones [12] or through net-carrying formations as demonstrated in the MIDRAS project [4], which forms the basis for the current IDAS project.

The effectiveness of UAV catching methods, particularly those employing autonomous interception systems, fundamentally depends on accurate target detection and localization capabilities. To achieve reliable autonomous interception, onboard detection systems capable of sub-meter precision are essential. An extensive survey of methods for drone detection and tracking has been conducted in [13].

Early UAV detection systems relied primarily on radar-based approaches, which face significant challenges in detecting small, low-flying targets with minimal radar cross-sections [14]. The integration of multiple sensors has emerged as the main solution for reliable UAV detection [15]. Multi-sensor fusion systems combine radar, electro-optical cameras, infrared sensors, radio frequency analyzers, and more to improve detection reliability [3,16]. However, most existing detection systems are designed for fixed stationary installation, providing target localization accuracy typically limited to several meters. This precision is insufficient for autonomous interception missions, where sub-meter accuracy is required for successful capture operations [17].

In particular, the integration of detection systems directly onto mobile interceptor platforms presents unique challenges, in comparison to ground-based systems. Limitations in weight, power consumption and computational resources significantly affect the overall capabilities. LiDARs have been used for autonomous navigation and obstacle avoidance in UAV applications, i.e., [18]. The comprehensive review by Ahmadi et al. [19] shows LiDAR as a fundamental sensor for detection systems. Lightweight LiDAR sensors, for example the LIVOX series [20], offering sufficient range and accuracy for the proposed

target detection while still meeting the required weight constraints, have become the main sensors for onboard detection.

Vision-based target recognition using machine learning approaches has demonstrated good effectiveness for UAV detection tasks [21,22]. YOLO-based architectures dominate real-time UAV detection, with recent variants achieving over 95% mAP while reducing model sizes by more than 50% [23,24]. Lightweight model optimization for embedded deployment further improves the usefulness in onboard detection [25]. YOLO version comparisons reveal that YOLOv3 and YOLOv5 perform best in air-to-air scenarios, while newer versions show reduced performance in these specific applications [26]. Deep learning models provide real-time object detection capabilities suitable for onboard processing on embedded systems [27,28]. Further event-based cameras on FPGA represent the cutting edge of low-latency processing [29]. For all vision-based systems, the development of specialized datasets of images of airborne UAVs, like UAVfly [30] and Det-Fly [31], is very important.

Ground-based detection, while functioning at large ranges, cannot provide the accuracy needed to intercept drones with a net in the air. The results shown in [13] consistently suffer higher estimation errors. Thus, air-to-air drone detection is employed for the final interception phase. The specific challenge of air-to-air or drone-to-drone detection represents a distinct and significantly more complex scenario compared to ground-based detection systems. Air-to-air UAV detection uses UAVs to carry onboard cameras and/or sensors to detect other UAVs in flight [30]. The dynamic flight characteristics of onboard cameras lead to significant variations in UAV appearance. Visual complexities arise as background elements, ground structures, and architectural features can obscure UAVs, further increasing the difficulty [26].

Despite significant progress, current air-to-air detection systems face several limitations. Most research focuses on UAV detection, without further localizing the target. Challenges remain in distinguishing between multiple overlapping drone images and handling the full spectrum of real-world operational environments [30]. The work presented in this paper contributes to addressing these gaps by providing onboard detection and localization capabilities specifically designed for autonomous UAV interception missions for the civilian use case.

3. Concept

This section presents the operational concept for the autonomous UAV interception system developed. The proposed system operates within the overall IDAS drone defense system [6], where multiple ground-based sensors, including radar, electro-optical cameras, and radio frequency analyzers, provide initial target detection and classification. As has been stated, the autonomous interception of unauthorized UAVs presents a complex challenge that requires seamless integration between the ground-based detection and the airborne interception systems.

The concept for the presented drone defense system consists of two distinct phases. The first phase uses the sensor data provided by the ground-based sensor network to guide the interceptor UAV to the general vicinity of the target. For the second or final interception phase, precision that exceeds the capabilities of ground-based tracking alone is required. To address this challenge, our approach introduces onboard detection capabilities that enable sub-meter precision during the final phases of the autonomous approach and catching of the target UAV. As soon as a possible threat is detected in the protected area, i.e., an airport, the interceptor system is commanded to start and guided to the approximate target location given by the stationary detection system. The first phase of guiding the interception UAVs to the target is illustrated in Figure 2.

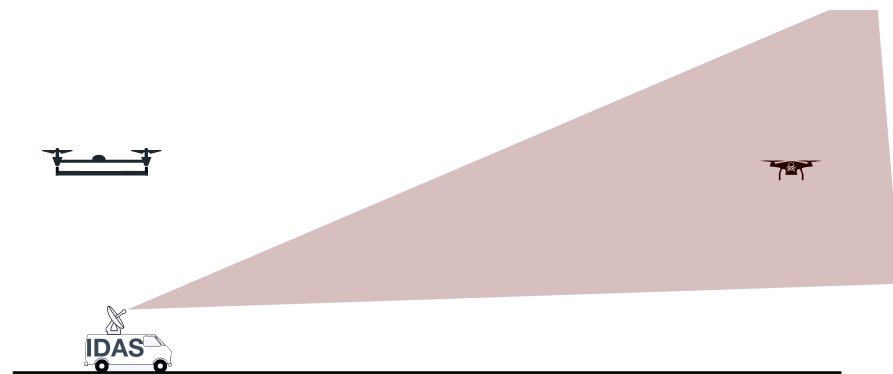


Figure 2. Initial target guidance phase.

The ground-based sensor fusion system provides target localization with typical accuracies of 5–10 m, which is sufficient for guiding the interceptor UAVs within the detection range of its onboard sensors. The transition to autonomous onboard detection is shown in Figure 3. Once the interceptor UAVs detects the target with its own onboard sensors, the system switches to an autonomous mode, relying on the integrated LiDAR and camera systems for target detection, tracking, and verification. Because the ground-based system pre-aligns the interceptor to fly directly toward the target before the onboard sensors take over, the target is kept naturally within the forward-facing field of view of both the detection sensors.

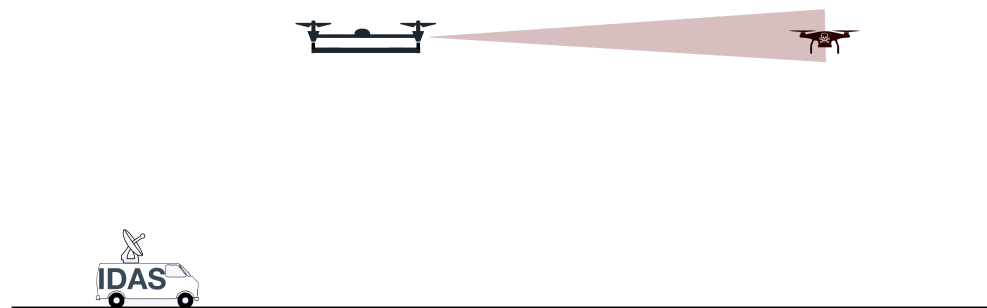


Figure 3. Autonomous detection phase.

The UAV uses two sensors for the onboard detection and verification of UAVs. The former is based on Light Detection And Ranging (LiDAR) data from a front-facing LIDAR Mid-70 featuring a 70 deg viewing angle sampled at a rate of 10 Hz. The latter uses image data from a front-facing global shutter Red–Green–Blue (RGB) camera e-CAM25_CUONX_H01R1 processed at 5 Hz. Both are mounted at the center of the UAV. The data generated by the sensors is processed in a receding manner as depicted in Figure 4. The loop is arranged as a 2×2 grid. The left column operates on detailed information; the right column uses higher-level abstractions. The top row handles sensing and cognition; the bottom row covers trajectory high-level planning and low-level control.

After the detection stage described in detail in the next chapter, the information on the target's position is then projected to its expected location in the camera frame. The image sampled from the camera is processed by a YOLO network retrained on images of UAVs which produces regions where UAVs are occurring in the camera frame. If the target lies within a region, it is verified.

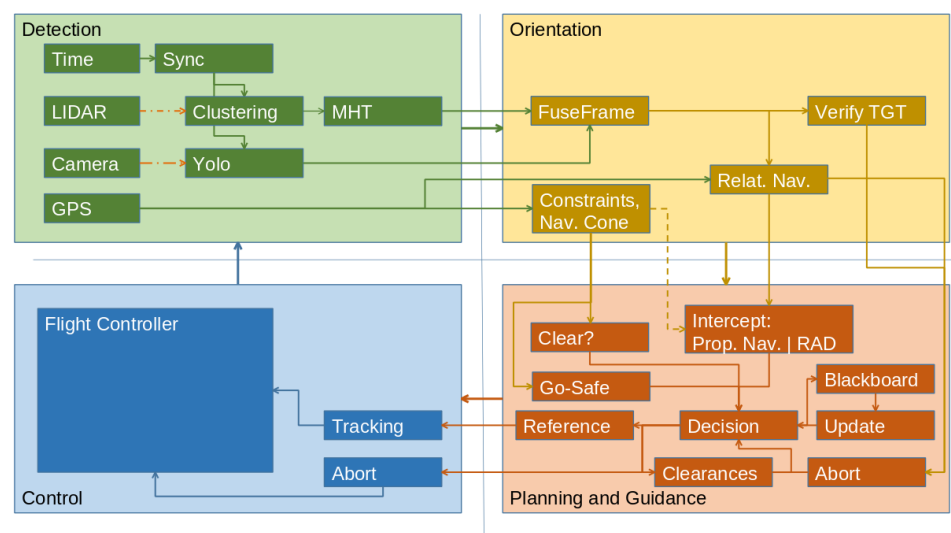


Figure 4. IDAS data processing loop.

Given the coordinates of the verified target, the planner then steers the UAV to intercept the target with the net. To achieve successful capture during this final phase, the system employs a Proportional Navigation (PN) guidance strategy. The PN guidance actively maneuvers the interceptor to maintain a constant line-of-sight rate to the target, naturally steering the UAV onto a collision course. The orientation in yaw is always controlled to look directly at the target. In the process, the desired pose for the controller is updated whenever new LiDAR data is available, closing the control loop. The high-frequency 500 Hz low-level control loop minimizes latency between detection updates and motor commands. The exact architecture and methods employed for detection and verification will be addressed in the next section.

4. Implementation

The act of targeting is decomposed into the detection of UAVs followed by their verification. Detection requires the positioning and labeling of UAVs given recurring measurement data. The problem itself is NP-hard and herein decomposed into subproblems which—albeit NP-hard as well—can be reasonably well solved with proven methods. The exact process is decomposed into the steps shown in detail in Figure 5 with their interactions.

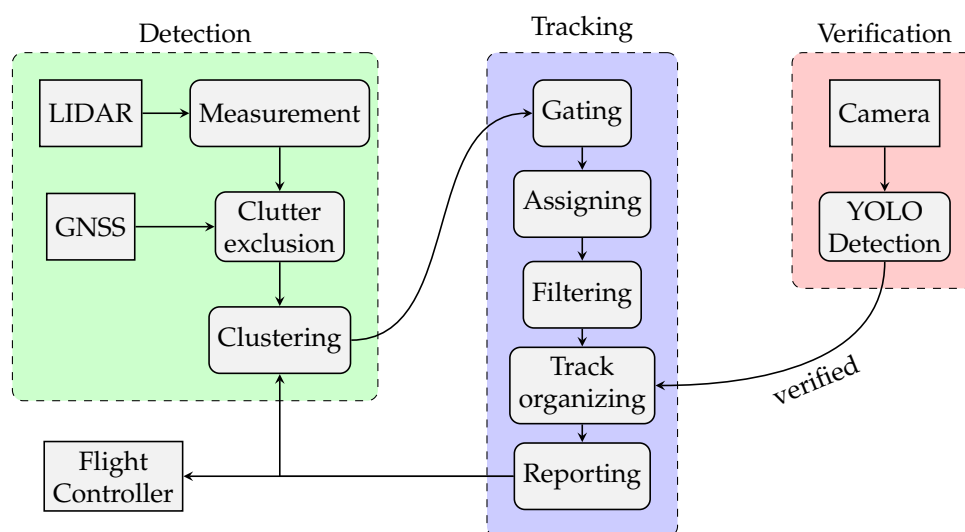


Figure 5. Overview of the implemented target detection, tracking and verification process.

4.1. Detection

The aim of the detection stage is to produce a point cloud wherein each element pertains to a cluster of points in the original cloud. It has to first exclude the ground and objects from the environment while ensuring that clusters originating from UAVs must not be ignored in the process.

Measurement is the first step towards detection. The collected data is processed by the Robot Operating System (ROS) driver, provided as a point cloud, and then converted into the UAV's coordinate system as well as the world coordinate system using the UAV's attitude estimation as well as Global Navigation Satellite System (GNSS) position. The opening angle of the LiDAR together with that of the camera is shown in Figure 6.

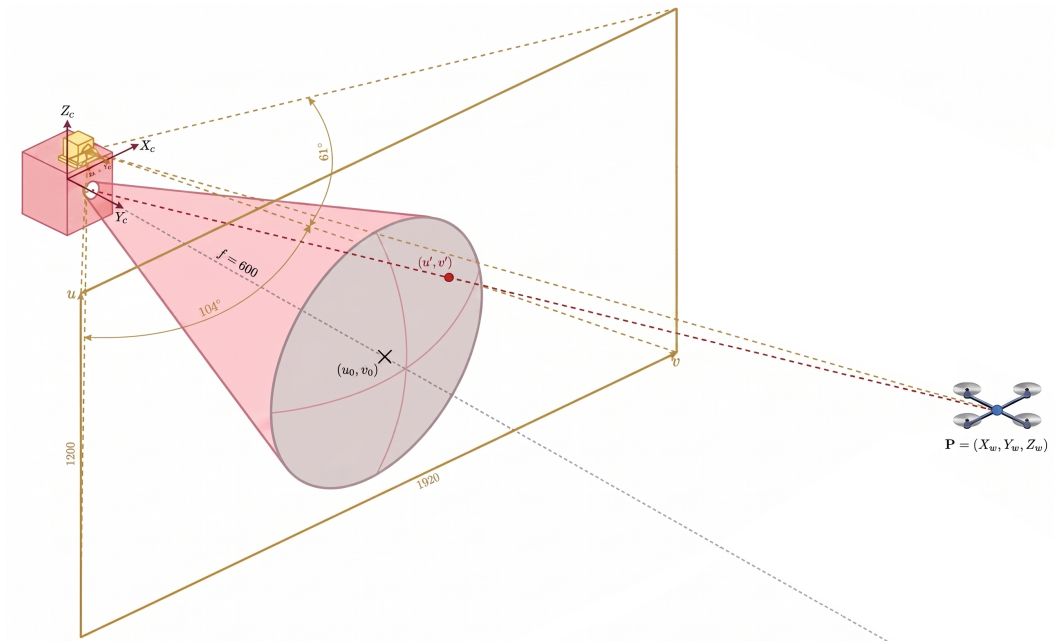


Figure 6. LiDAR and camera detection cones.

Clutter Exclusion removes any points likely due to the environment. This serves primarily to reduce the computational load in later parts of the guidance system. The filtering is carried out in three steps:

First, the box filter includes all points that are located in a box in relation to the UAV. Then, the minimum height filter includes all points that are above a preset minimum height. And finally, the voxel filter removes points that are too close to each other. The filters are based on the *Point Cloud Library* [32].

Field tests have shown that this filter chain efficiently filters out the environment and only the target UAV is still included in the resulting point cloud. A point cloud of several thousand points, consisting mostly of interference signals, is thus transformed into a point cloud of 4 to 30 points with isolated interference signals.

Clustering is the final step towards target acquisition which marks locations in the point cloud where a UAV might be. To do this, we assume that each cluster in the point cloud is based on a UAV. This assumption is later verified in the target verification via the vision-based neural network pipeline, detailed in Section 4.3. Tracking the last target position for target detection increases sensitivity. On the one hand, if the target is far away or has not yet been detected, a ring buffer containing up to ten point clouds is maintained. On the other hand, points near the last estimated position are weighted more heavily.

Cluster detection is implemented using DBSCAN's [33] "EuclideanClusterExtraction". For our real-world flight tests, the clustering parameters were tuned to a maximum cluster

distance threshold (ϵ) of 0.3 m and a minimum cluster size of 4 points. Let n be the number of points in the filtered point cloud. The algorithm used has an expected runtime of $O(n)$ to identify clusters. In fact, we observe that the entire filtering process only uses a fraction of the available computing power. The clusters are then passed on as a point cloud for target tracking.

4.2. Tracking

The clusters are then tracked with the aim of assigning persistent labels and estimating their velocities as well as their position in space by fusing measurements over time. Target acquisition itself can detect multiple clusters, which raises the question of which cluster is actually based on the previous history of clusters or is simply a measurement error, and how to determine how many UAVs have been detected and where they are moving. These problems are addressed by the tracking module.

To understand target tracking, it is essential to distinguish between clusters and trajectories. A cluster is a potential target that has been detected in target acquisition. However, it is neither verified, identified, nor described in more detail than as a point in space. A trajectory is a hypothesis that a UAV is currently located at a specific location. This UAV is tracked in the sense that it is assigned a number, and its speed and intent are further estimated. Target tracking maintains a list of up to eight trajectories, which it updates iteratively based on the clusters so that the distance between trajectories and clusters is minimal. This problem is also NP-hard, but is solved efficiently in practice.

Gating is the first step. Due to the available computing power, the UAV can track a maximum of eight flight paths, but more than eight clusters can be detected. Gating removes clusters that are far from the existing flight paths until only eight clusters remain.

Assigning then associates the remaining clusters with existing or possibly new trajectories. To do this, IDAS uses the Global Nearest Neighbor (GNN) algorithm [34], as it is simple and meets the requirements at the same time. For each trajectory, the Mahalanobis distance between each cluster is determined. Subsequently, each trajectory is assigned a cluster using the Hungarian method [35].

Filtering fuses the history of measurements to form a complete estimate of the target's state. While GNN takes care of associating clusters and organizing trajectories, the filter's task is to estimate trajectories using associated clusters, referred to here as measurement points. To do this, IDAS uses optimal estimation methods, in particular Moving Horizon Estimation (MHE). MHE is easier to understand from a perspective that first considers Full Information Estimation (FIE). FIE estimates the current state of the UAV, consisting of position and velocity, using all past measurement points. Since the resulting optimization problem grows over time, MHE considers the last N measurement points, where N is called the horizon length (where $N = 5$ during our real-world flight evaluation). The greater weighting of the last known measurement point is taken into account by means of the so-called arrival cost. IDAS's MHE has been modified to be robust against outliers. The standard formulation of Moving Horizon Estimation (MHE) uses a pure quadratic cost function. Outliers are weighted more heavily than actual measurement data in this formulation. The MHE from IDAS uses a Huber cost function [36], which starts quadratically and transitions to a linear function. This means that outliers are weighted less heavily, which improves the quality of the estimation. By virtue of this filtering, predictions of the target's trajectory can be made which in turn eases the association and increases tracking reliability.

Track organization manages the flight paths and marks confirmed flight paths, as well as flight paths that are pending deletion because there were no more clusters associated with them. The flight paths are then passed on to the target selection system.

Reporting is the last step which given the tracks as well as a target selection reports the target's 3D position to the high-level planner to give the respective commands to the low-level flight controller acting on the motors of the UAV.

4.3. Verification

Once a target has been detected and trajectory guidance is active, it can be selected. The selected target is verified before it is captured. This is done using artificial intelligence, which evaluates image data and classifies it as either a UAV or something else. An example of detection using a neural network is shown in Figure 7.



Figure 7. Detection of UAVs with You Only Look Once (YOLO).

The image data is captured with a global shutter camera, which prevents smearing or overlapping. The viewing angle is shown in Figure 6 in comparison to the opening angle of the LiDAR. A custom neural network based on the highly optimized YOLOv7-tiny architecture [24] is then used to detect UAVs in the image via bounding-box regression. This specific lightweight architecture was selected because it is optimized for edge-GPU deployment, achieving rapid inference times of approximately 30 ms onboard the Jetson Orin Nano, optimal for real-time verification. To ensure robust recognition across various backgrounds and drone models, the network was trained using a publicly available dataset for drones (hosted via Roboflow [37]) and for further training a new dataset was created containing labeled images of a wide range of UAVs from video data.

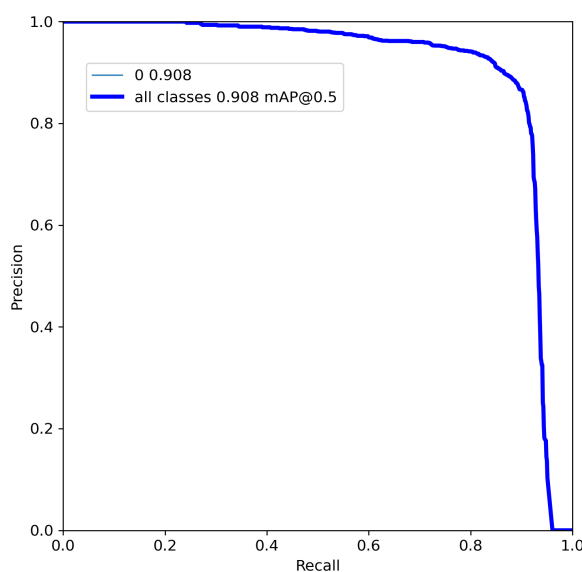


Figure 8. Precision–Recall curve YOLO.

The precision–recall curve is shown in Figure 8 and shows a high area under the curve performance. In the next step, the image is used to calculate the direction of the detected UAV in world coordinates so that this direction can be compared with the detection by

LiDAR. The most common method in robotics is to model the camera using its distortions and a transformation matrix from world coordinates to image coordinates. The extrinsic parameters are already taken into account by the coordinate transformations. To determine distortion and the intrinsic matrix, calibration is necessary, for which images of a known pattern, shown in Figure 9, were taken from different angles. Based on the target guidance, the system calculates where a UAV should be visible in the image. If a UAV is indeed located there in the image, the target is verified and the capture routines are cleared to fly towards it.

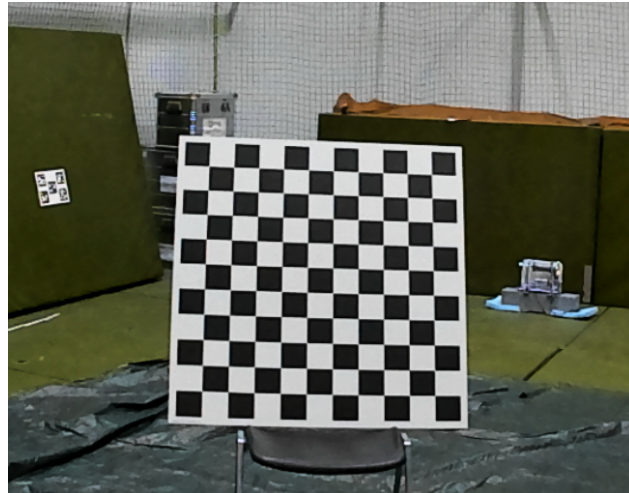


Figure 9. Checker board camera calibration.

5. System Design

The detection and verification system is integrated into a custom-built octocopter platform designed specifically for the autonomous drone interception missions in IDAS. The interception UAV has a dimension of 2×1 m and is designed to carry a maximum load of 5 kg. A CAD model of the UAV can be seen in Figure 10.

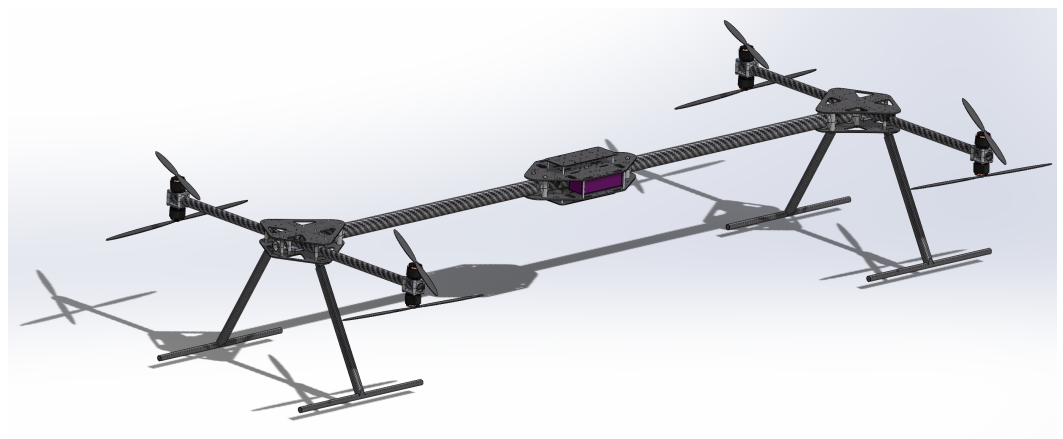


Figure 10. CAD model of the interception UAV.

The components used for the real-world demonstration and later discussed evaluation of the system can be seen in Figure 11. Here, the key components used for the detection and verification of the target are highlighted.

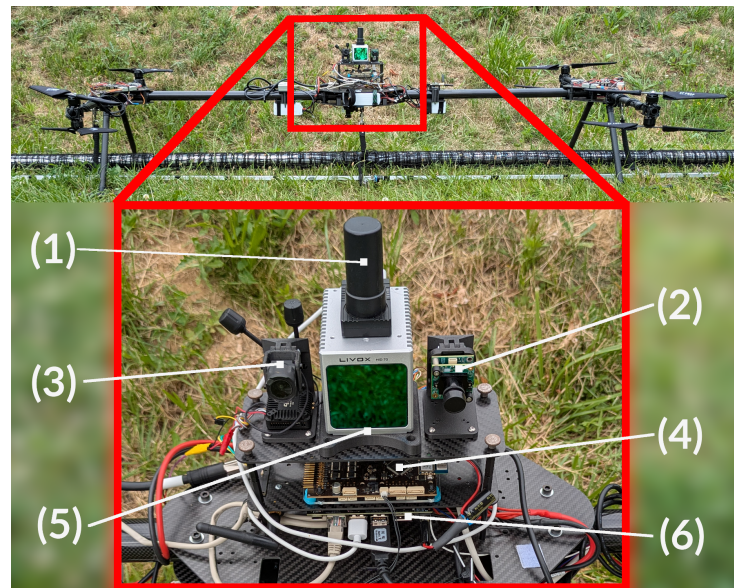


Figure 11. Hardware configuration of the detection system mounted on the interceptor UAV.

5.1. Detection Hardware Components

The core detection system consists of several Components-Off-The-Shelf (COTS). The primary detection sensor (5) is a LIVOX Mid-70 LiDAR unit [20], which provides 3D point cloud data with a 70° field of view and 150 m operational range. The sensor weighs 760 g and consumes 8 W of power, making it suitable for integration on UAV platforms. The LiDAR operates at up to 100,000 points per second with 2 cm ranging accuracy. This enables target localization in 3D space at high precision.

For target verification (2), an e-CAM25_CUONX_H01R1 global shutter camera from e-con Systems [38] is used. This global shutter camera eliminates rolling shutter artifacts during high-speed maneuvers, which is important for accurate target identification during dynamic flight operations. The camera interfaces directly with the companion computer via MIPI CSI-2, providing real-time video streams for neural network processing.

The computational backbone of the system (6) is an NVIDIA Jetson Orin Nano, which serves as the companion computer for all detection and verification algorithms. The Jetson processes the LiDAR point clouds and the camera imagery in real time running Ubuntu 20.04 with ROS Noetic.

5.2. Supporting Platform Components

High-position-accuracy data is provided through a dual-frequency Holybro F9P GNSS receiver (1), which is essential for coordinate frame transformations between the global reference frame data of the target, given by the ground-based detection system and the onboard detection sensors used for the final autonomous approach and catch.

Flight control is managed by a custom-manufactured flight controller PCB (4) designed specifically for this application at the department in [39].

The flight controller hardware integrates automotive-grade IMU sensors, a magnetometer, and a barometer for robust attitude estimation. The control architecture controlling the UAV during the interception flight is described in detail in [40] and runs at a 500 Hz control loop frequency.

An additional DJI Airunit FPV camera (3) provides a real-time video feed to the ground control station for the operators, i.e., law enforcement, though it is not directly involved in the autonomous detection algorithms.

5.3. Data Flow Architecture

The detection system operates through a distributed processing architecture where sensor data flows through multiple processing stages. Raw LiDAR point clouds from the LIVOX sensor are transmitted via Ethernet to the Jetson Orin Nano, where they are processed, as described in the previous chapter.

Simultaneously, the verification camera streams image data to the Jetson for processing through the custom-trained YOLO neural network in real time. Communication between the Jetson companion computer and the flight controller occurs via high-speed UART at 500 Hz.

6. Evaluation

The performance of the integrated LiDAR-based detection and vision-based verification system is of course critical for the success of the proposed autonomous drone interception system. To validate the effectiveness of the developed approach, comprehensive evaluations were conducted under both controlled laboratory conditions and realistic scenarios. First, the visual-based verification system was evaluated. For this the interceptor drones detected and verified different types of UAVs in the air. Next, the combined LiDAR- and camera-based system was evaluated in a stationary test, where only the target was moving, before finally looking at a real-world interception of a target. This evaluation used high-precision Real-Time Kinematic GNSS (RTK-GNSS) as the ground-truth reference to evaluate the detection accuracy.

6.1. Verification System

After the LiDAR-based detection and localization algorithms identify potential target clusters, these targets must be verified using computer vision to ensure reliable identification before engaging the final catch routine. Figure 12 shows the real-time output from the verification camera during a test flight, with bounding boxes clearly identifying the detected target. Figure 13 provides a third-person perspective of the same verification scenario, illustrating the operational context in which the visual verification system operates.

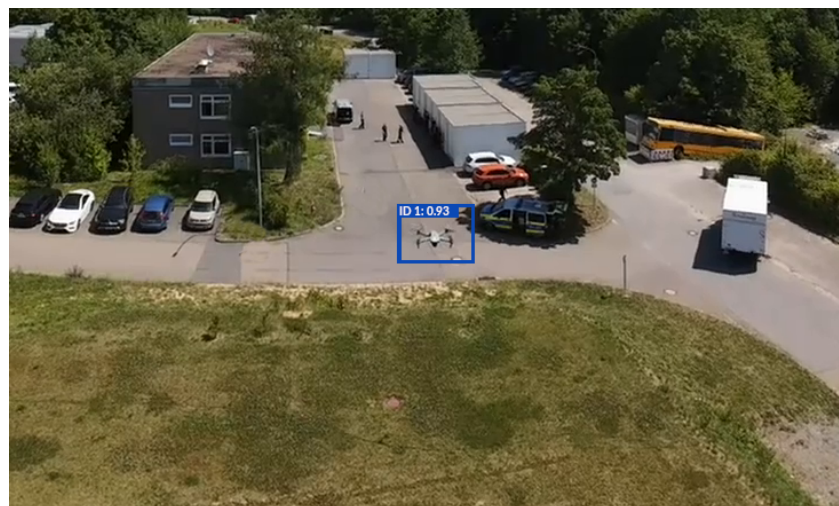


Figure 12. Verification camera view with bounding box overlay.

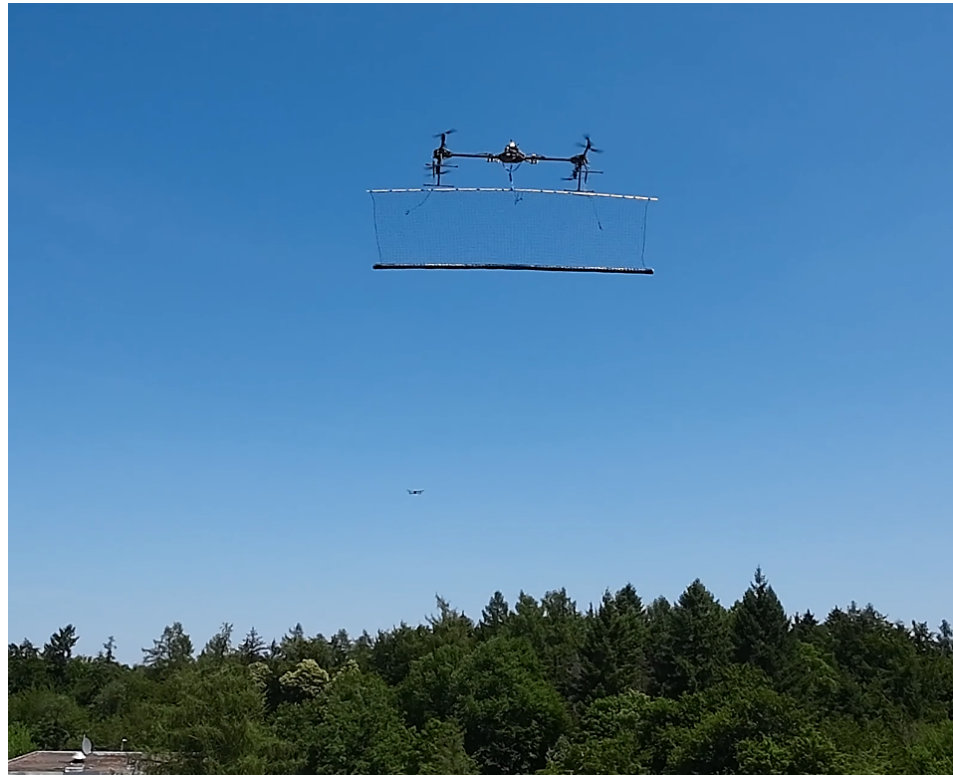


Figure 13. Third-person view at same timestamp.

To validate the system's reliability, more than 100 autonomous flight attempts were conducted. Out of these flights, the system successfully intercepted the target in over 90% of cases (fewer than 10 misses). The optimized YOLOv7-tiny network demonstrated highly robust verification performance, achieving a Mean Average Precision (mAP@0.5) of 0.96. During validation, the model reached a terminal precision of 0.89 and a recall of 0.88, yielding an F1 score of 0.88. While the system successfully detected and verified several different types of UAVs, all current real-world flight tests were conducted under clear daylight conditions. Consequently, the vision-based verification's performance in low-light scenarios or adverse weather conditions remains untested and will be the subject of future development.

6.2. Combined Detection

To evaluate the accuracy and reliability of the combined LiDAR detection and visual verification system, static evaluation tests were done using high-precision RTK-GNSS as the ground-truth reference. In the following evaluation, the sensor suite (LIVOX mid-70 LiDAR sensor, global shutter verification camera, flight controller and GNSS) were mounted on a tripod in a stationary configuration. A target drone equipped with RTK-GNSS providing centimeter-level positioning accuracy was flown through various flight patterns within the detection range. Here, and also for the later evaluations in flight, the RTK-GNSS system served exclusively as the ground truth for post-analysis and was not used for flight control or detection algorithms. Results demonstrate that the onboard detection system reliably identifies and tracks target drones with sub-meter accuracy, as shown in Figure 14.

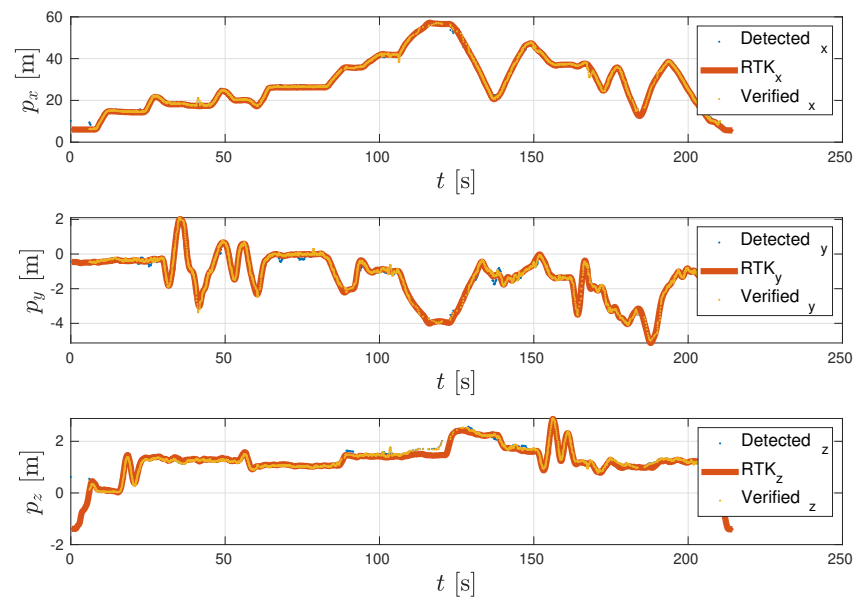


Figure 14. Static evaluation results showing detection accuracy compared to RTK-GNSS ground truth.

During extensive real-world testing, the system successfully detected and intercepted several different types of UAVs, with target velocities up to 5 m/s. The target drone used for the evaluation in this paper had dimensions of $420 \times 420 \times 200$ mm ($L \times W \times H$) and was equipped with an RTK-GNSS module used exclusively to establish the ground-truth data. This specific target was detected at a maximum range of 60 m. Of course, the accuracy of the detection is better at closer distances, and targets with a bigger footprint would be detected at even further distances. Further, the verification of the target does not always work at the furthest distances available through the detection process, but works consistently at distances of <30 m. The error characteristics are shown in Table 1.

Table 1. Detection and verification accuracy statistics compared to RTK-GNSS ground truth.

Measurement Type	Axis	Mean [m]	Std [m]	RMS [m]
Detection Only	X	0.2744	1.2479	1.2773
	Y	−0.0301	0.2949	0.2963
	Z	−0.0022	0.1044	0.1044
Verified Detection	X	0.2408	0.7193	0.7585
	Y	0.0062	0.1462	0.1463
	Z	−0.0045	0.0800	0.0801
Verified Close Range	X	0.2890	0.5061	0.5827
	Y	0.0108	0.1247	0.1252
	Z	−0.0420	0.0562	0.0701

The results show the improvement in detection accuracy through the implemented verification process. While the LiDAR-only baseline already achieves good results, the integration of visual verification significantly enhances overall system performance by reducing positioning error. Especially in the closer range, the implemented detection system shows very promising results. The lower accuracy in the x-axis results from an offset of the ground-truth GNSS-antenna mounted in the middle of the detected UAV, while the detection algorithms provide the closest points in view. This means that the mean error in all axes is below 10 cm.

6.3. Overall System

The final validation of the performance of the implemented detection system was conducted through real-world interception flights, where both interceptor and target drones were equipped with RTK-GNSS systems to provide ground-truth positioning data. Figure 15 shows a complete interception maneuver. The target is tracked throughout the entire engagement sequence. The data shows the interceptor (blue) successfully approaching and intercepting a target drone (orange) from a distance exceeding 40 m, while the onboard detection system (yellow) continuously tracks the target position throughout the approach. This shows the performance in a dynamic situation, where the interceptor UAV is flying under realistic operational conditions.

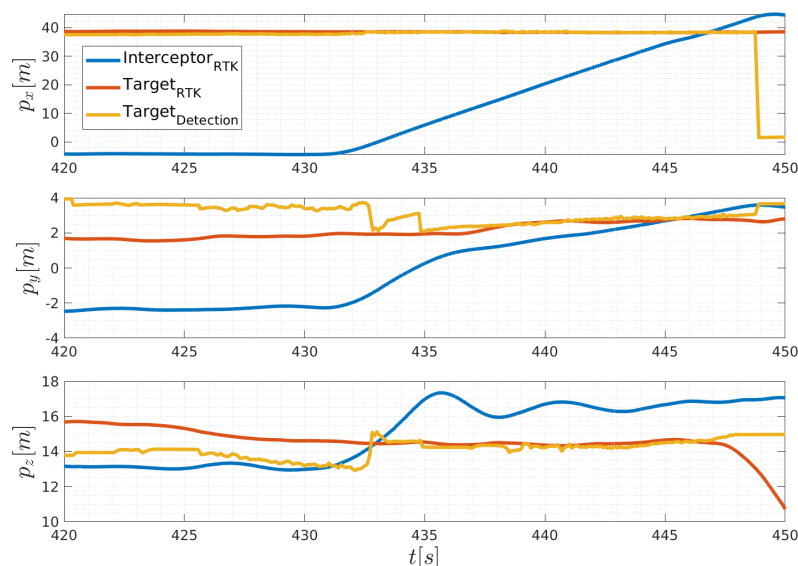


Figure 15. Position of the UAVs during a real-world catch.

Figure 16 provides detailed analysis of the detection accuracy during the final interception phase at ranges below 30 m, where positioning precision becomes most critical. Here, the error stays well within the required tolerances of the net size of 4×4 m, as it is below 0.4 m in all axes. To validate the practical reliability of the system beyond single-flight metrics, more than 100 autonomous interception flight attempts were conducted over the course of the real-world testing. Out of these flights, fewer than 10 missed the target, resulting in an overall real-world interception success rate exceeding 90%.

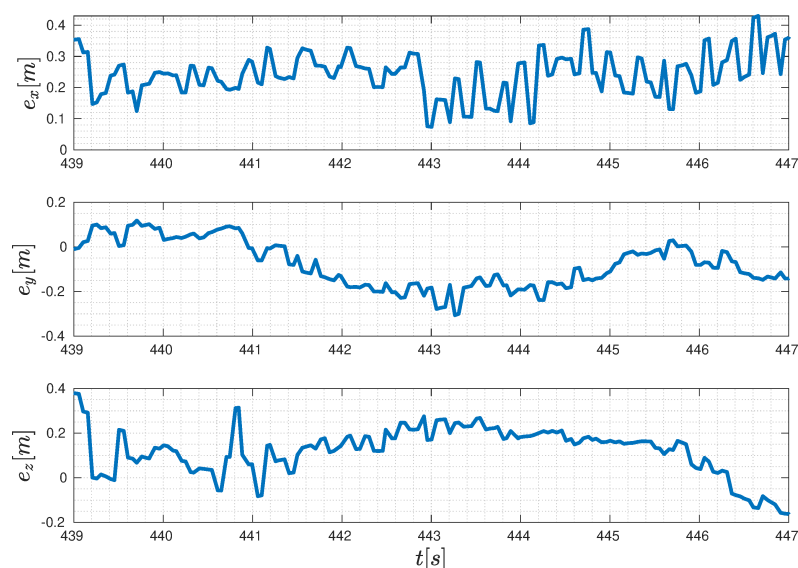


Figure 16. Detection error during a real-world catch.

7. Conclusions

This paper presents a LiDAR-based detection and vision-based verification solution for an autonomous drone interception system for the project IDAS, where the interception UAV carries a net to catch potentially dangerous UAVs in the air autonomously. The system successfully addresses the critical challenge of achieving sub-meter precision during the final interception phase, where ground-based sensors alone prove insufficient for reliable target capture. Rather than purely theoretical algorithmic novelty, the core contribution of this work is the applied systems engineering and real-world integration that enables dynamic interception. The evaluation showed that the system has up to 60 m range with sub-meter accuracy, while vision-based verification achieved over 90% detection rates at ranges below 30 m. Real-world interception tests confirmed the system maintains positioning accuracy well within the required tolerances, with detection errors below 0.4 m in all axes during the critical final approach phase. The experimental results validate the effectiveness of the dual-sensor approach, where the LiDAR provides the target localization and tracking, while camera-based verification reduces false positives before starting the final capture routines.

Despite these positive outcomes, several limitations remain for future work. As the system is just a prototype at *TRL level 6–7*, several assumptions were made. All flights for the evaluations were conducted under clear daylight conditions. Consequently, the vision-based verification's robustness under bad weather conditions (e.g., rain or fog), varying illumination, and heavy target occlusion remains untested. Furthermore, the current operational envelope was evaluated using standard commercial UAVs flying at speeds up to 5 m/s. Intercepting highly agile targets, such as fast-flying racing drones or fixed-wing UAVs, falls outside the current system boundaries and represents a crucial area for future development.

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Abbreviations

The following abbreviations are used in this manuscript:

COTS	Components-Off-The-Shelf
FIE	Full Information Estimation
GNN	Global Nearest Neighbor
GNSS	Global Navigation Satellite System
IDAS	Innovatives Drohnenabwehrsystem (Engl. Innovative Drone Defense System)
LiDAR	Light Detection And Ranging
MIDRAS	Micro Drone Protection System
MHE	Moving Horizon Estimation
RGB	Red–Green–Blue
ROS	Robot Operating System

RTK-GNSS	Real-Time Kinematic GNSS
UAV	Unmanned Aerial Vehicle
YOLO	You Only Look Once

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